The Source Sector

What Orbits Identify, How a Bilocal Field Gravitates, and When It Does Not

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Companion to "Least-Cost Holonomy and Co-Distinction Fields: The Shields Field of Distinct Actions". That paper defined the field and proved when it admits a potential. This one asks what it takes for such a field to source gravity, and finds that the answer is not always.

The paper is in two halves separated by a printed line. Parts One to Five use standard general relativity and field theory, and stand or fall on that. Part Six is an extension in the author's own framework, Unified Mechanics, and is offered as interpretation. Nothing above the line depends on anything below it.

A theory that supplies a source must say what the source is made of, in what units, and whether it can be measured. Two of those three have answers here. The third has a theorem saying it does not.

Abstract

Two results, one negative and one restrictive.

The first is an identifiability theorem. Newtonian and relativistic two-body motion depends on the gravitational coupling and the source mass only through the product $\mu = GM$, so the transformation $G \to \lambda G$ with $M \to M/\lambda$ leaves every trajectory invariant. No quantity of orbital data, from any number of systems, separates the universal coupling from the absolute source normalisation. Orbits identify a geometric source strength and not kilograms. This is stated as a theorem rather than a caution because it sets what a field theory of the source sector may be asked to predict: μ, or GM/c^2, or the full stress-energy tensor, but not the SI value of G from orbital data alone.

The second concerns how a bilocal field can gravitate at all. The Shields field $\mathcal{D}(x,y)$ is defined on pairs, and general relativity is sourced locally, so $\mathcal{D}$ cannot be inserted into the stress-energy tensor as though it were a scalar at a point. Representing local transport by a one-form and building a Maxwell-type action from its curvature is the natural route, and it carries an exact restriction:

Proposition. If $\mathcal{D}$ is locally potentiated, the associated connection is flat, its curvature vanishes identically, and the stress-energy tensor built from that curvature is zero. A locally potentiated co-distinction is gravitationally inert in the connection representation, and its obstruction, where one exists, is purely topological.

The consequence is a criterion, and the criterion is then computed rather than left open. In the diffusion closure of transport theory the curvature of the transport one-form is $\nabla\times\omega = (\nabla\kappa_t \times \nabla u_r)/(3\kappa_t^2 u_r)$, so the sector supplies a local source if and only if the opacity gradient and the radiation-energy gradient are non-parallel. A homogeneous scattering medium is flat and

gravitationally inert however optically thick it is. This is more restrictive than the expectation it replaces and it corrects it: scattering supplies the mechanism, misalignment supplies the curvature.

Two further items the earlier draft listed as tasks are closed here rather than deferred. Covariant conservation of the source is automatic by Noether second theorem for any diffeomorphism-invariant action and is not a per-model check. And the identification of the background scalar with the Newtonian potential holds at first post-Newtonian order with fractional corrections of order Φ_N/c^2, which is 10^{-8} at Earth orbit and therefore far below the precision of the maps that use it, while failing outright in strong fields. What remains genuinely open is the prediction of μ from closure structure, which Theorem 1.1 shows is required before any orbital data can test the sector at all.

A candidate covariant action, its stress-energy tensor, the weak-field bridge and a falsification programme are set out on that basis.

Keywords: identifiability; geometrised mass; stress-energy tensor; bilocal source; flat connection; weak-field limit; falsification.

PART I · WHAT ORBITS IDENTIFY

1. The μ degeneracy

Gravity enters two-body acceleration through a single product:

$$g = -\mu\, r / |r|^3 = -GM\, r / |r|^3, \qquad \mu := GM.$$

Theorem 1.1 (μ-identifiability). Let the trajectory of a test body solve $\ddot{r} = -GM\, r/|r|^3$. Then every functional of the trajectory depends on the pair (G, M) only through μ = GM. Consequently no set of orbital observations, of any size or precision, identifies G or M separately.

Proof. The equation of motion contains G and M only in the combination GM. Two parameter pairs (G,M) and (λG, M/λ) therefore generate identical solutions from identical initial data, for every $\lambda > 0$. Any functional of the resulting trajectories takes the same value on both. ■

Increasing the number of planetary or exoplanetary systems improves the estimate of the common law and its scatter. It does not and cannot separate the universal coupling from the absolute source normalisation, because the two enter through one number.

It is worth being precise about what kind of failure this is, because the companion papers of this programme distinguish two kinds and the distinction matters here. This is a redundancy, not a loss. The map $(G,M) \mapsto \mu$ has fibres that are stable under the dynamics, since the evolution depends only on μ, so the quotient descends and nothing dynamically relevant is discarded. It is the gauge case, not the boundary-term case. μ is a good variable precisely because the projection to it is safe.

Orbits do not fail to measure G. There is nothing in an orbit that G names.

2. What a source theory may therefore be asked for

The geometrised source is the natural intermediate target:

$$m_{geom} = GM/c^2 = \mu/c^2, \qquad r_s = 2GM/c^2.$$

A field theory can predict μ, or GM/c^2, or the full stress-energy tensor, without expressing its raw amplitude in kilograms. Kilograms enter only with a choice of SI units, or when one attempts the separate and much harder task of deriving the numerical SI value of G.

Programme mode	Imported	The field must determine
General relativity as toolkit	measured G and the Einstein equation	a covariant source $T_{\mu\nu}$ from φ, ψ, $\mathcal{D}$ and any active fields
Geometric prediction	G implicit in geometrised units	the metric source scale GM/c^2 or an equivalent invariant
Derivation of G	only non-gravitational standards, c and possibly $\hbar$	both an absolute inertial-energy scale and the gravitational source strength

Only the third mode yields a non-circular prediction of the SI coupling, and only the third requires an independent inertial-energy standard. The first two are achievable without one, and this paper works in the first.

PART II · GENERAL RELATIVITY AS THE TOOLKIT

3. The division of labour

The architecture is not to replace general relativity but to supply it a source. Einstein geometry performs the covariant conversion:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4)\, T_{\mu\nu}^{UF}.$$

The left side is standard. The right side is where the proposed objects must earn their interpretation, and the central task is to write a diffeomorphism-invariant matter action and vary it with respect to the metric:

$$T_{\mu\nu}^{UF} = -(2/\sqrt{-g})\, \delta S_{UF} / \delta g^{\mu\nu}, \qquad \text{with} \qquad \nabla^\mu T_{\mu\nu}^{UF} = 0 \text{ on solutions.}$$

For driven or interrogative dynamics individual sectors may exchange energy and momentum, so the passive and active tensors need not be separately conserved provided the total is: $\nabla^\mu T^{passive}_{\mu\nu} = Q_\nu$ and $\nabla^\mu T^{active}_{\mu\nu} = -Q_\nu$.

4. A candidate local action

In the sector where the co-distinction is exact and admits a scalar potential φ, a minimal local action can be written. This is a candidate model and not a derivation from the framework's axioms. Signature $(-,+,+,+)$.

$$S = (c^4/16\pi G) \int d^4x \sqrt{-g}\, (R - 2\Lambda) + \int d^4x \sqrt{-g}\, \mathcal{L}_{UF},$$

$$\mathcal{L}_{UF} = -\tfrac{1}{2}K(\varphi)\partial_\mu\varphi\partial^\mu\varphi - V(\varphi) - (D_\mu\psi)^* D^\mu\psi - U(|\psi|,\varphi) - \tfrac{1}{4}Z(\varphi)F_{\mu\nu}F^{\mu\nu} + \mathcal{L}_{active}.$$

K controls scalar stiffness, V the background and closure-supporting potential, $\psi = R\, e^{i\theta}$ carries local cyclic closures, $D_\mu = \nabla_\mu - iqA_\mu$ is an optional gauge-covariant derivative, U couples closure amplitude to the background, Z permits a field-dependent radiative coupling, and $\mathcal{L}_{active}$ carries the degrees of freedom needed for driven growth or fission.

5. The stress-energy tensor

Varying the passive terms gives standard structures:

$$T_{\mu\nu}^{(\varphi)} = K(\varphi)\, \nabla_\mu\varphi\, \nabla_\nu\varphi - g_{\mu\nu}[\tfrac{1}{2}K(\varphi)(\nabla\varphi)^2 + V(\varphi)],$$

$$T_{\mu\nu}^{(\psi)} = (D_\mu\psi)^* D_\nu\psi + (D_\nu\psi)^* D_\mu\psi - g_{\mu\nu}[(D_\alpha\psi)^* D^\alpha\psi + U(|\psi|,\varphi)],$$

$$T_{\mu\nu}^{(rad)} = Z(\varphi)[F_{\mu\alpha} F_{\nu}{}^{\alpha} - \tfrac{1}{4}g_{\mu\nu} F_{\alpha\beta}F^{\alpha\beta}].$$

The tensor is not chosen separately from the action. The action fixes both the equations of motion and the source supplied to Einstein geometry, and that is the discipline the whole construction depends on.

Proposition 5.1 (Conservation is automatic). If S_{UF} is a diffeomorphism-invariant functional of the metric and the matter fields, then the divergence of $T_{\mu\nu}$ vanishes identically on solutions of the matter equations of motion, with no further condition on K, V, U, Z or the active sector.

Proof. Under an infinitesimal diffeomorphism generated by ξ the metric varies by $\nabla^\mu\xi^\nu + \nabla^\nu\xi^\mu$ and the matter fields by their Lie derivatives. Invariance of S_{UF} gives zero for the total variation, and on shell the matter contribution vanishes. Substituting the definition of $T_{\mu\nu}$ and integrating by parts leaves the integral of $\xi_\nu \nabla_\mu T^{\mu\nu}$ against arbitrary compactly supported ξ, hence $\nabla_\mu T^{\mu\nu} = 0$. This is Noether second theorem. ■

This closes what the earlier draft listed as a task to be verified. Covariant conservation is not a property to be checked case by case once a Lagrangian is chosen. It follows from writing the source as a metric variation of a diffeomorphism-invariant action, which Section 3 already required. The only way to lose it is to write $T_{\mu\nu}$ by hand instead of deriving it, and Section 5 does not do that.

6. Where the mass comes from

A localised closure needs no amplitude labelled in kilograms. Once the action has physical dimensions the configuration carries an energy density through T_{00}, and its effective gravitational source is the integral of that density including amplitude, gradients, phase motion, potential energy, gauge energy and active resources:

$$\rho\_eff = T_{00}\text{^}UF/c^2, \qquad M\_eff = \int\_\Sigma \rho\_eff \sqrt{\gamma}\, d^3x.$$

The amplitude normalisation and the Lagrangian together establish energy units. The theory does not declare that ψ is measured in kilograms; it must specify an action whose finite-energy closures possess a calculable T_{00}.

PART III · THE BILOCAL PROBLEM

7. Why an unpotentiated field is not a local source

The Shields field $\mathcal{D}$(x,y) is defined on ordered pairs. General relativity is sourced locally, by a tensor density at a point. An unpotentiated co-distinction therefore cannot be inserted into T_μν as though it were a scalar at x, and no amount of notation removes the difficulty: there is no value at x to insert.

Two routes are available and they are not equivalent.

Route	Form	Requirement
Local connection	represent local transport by a one-form B_μ with curvature H_μν = ∇_μB_ν − ∇_νB_μ; build a covariant action from H	the connection must faithfully represent the measured co-distinction
Explicit bilocal	S_$\mathcal{D}$ = ½∫∫ $d^4x\, d^4y$ √−g(x)√−g(y) W(x,y)$\mathcal{D}$(x,y)^2	a diffeomorphism-covariant kernel W, and acceptance of a nonlocal source

The next section shows that the first route is not a matter of convenience. It is forced where it applies, and empty where it does not.

8. The flatness proposition

The companion paper defines the one-form associated with a smooth co-distinction as ω := d_x$\mathcal{D}$(x,y)|_{y=x}, and this is the canonical candidate for the connection B of route one. Its curvature is H = dω.

Proposition 8.1 (Flatness of a locally potentiated field). Let $\mathcal{D}$ be smooth and locally potentiated on M, meaning that every point has a neighbourhood U carrying L_U with $\mathcal{D}$(x,y) = L_U(x) − L_U(y) for all x, y in U. Then $H = d\omega$ vanishes identically on M, and any local action built from H alone vanishes, so the associated stress-energy tensor is identically zero.

Proof. On U, $\omega = d_x(L_U(x) - L_U(y))|_{y=x} = dL_U$. Hence $H = d\omega = d^2L_U = 0$ on U. Since the point was arbitrary, $H \equiv 0$ on M. A Lagrangian of the form $-\frac{1}{4}Z\, H_{\mu\nu}H^{\mu\nu}$ then vanishes pointwise, and its metric variation vanishes with it. ■

A locally potentiated co-distinction is gravitationally inert. Its obstruction, where one exists, is purely topological, and topology does not curve spacetime.

Corollary 8.2 (Source criterion). The co-distinction sector supplies a non-vanishing local source in the connection representation if and only if $H = d\omega \neq 0$ pointwise, equivalently if and only if the cocycle condition fails on arbitrarily small triples. Global failure alone is not sufficient.

This changes the status of the two routes. Where $\mathcal{D}$ is locally a difference and only globally not, route one is available and returns nothing, and route two is the only representation that carries the obstruction at all. Where $\mathcal{D}$ fails the cocycle at every scale, route one is not merely convenient but canonical, because ω is determined by $\mathcal{D}$ rather than chosen.

> Proposition 8.1 and Corollary 8.2 are PROVED. They are statements about any co-distinction field and do not depend on the framework.

9. The curvature computed for a named kernel

Corollary 8.2 turns the question into a calculation, and the calculation can be done in the standard closure of transport theory rather than left as a programme item. Take the diffusion approximation, valid when scattering dominates absorption, in which the flux obeys Fick law with total transport opacity κ_t:

$$F = -(c / 3\kappa_t)\ \nabla u_r.$$

The natural transport one-form is the normalised flux, which is the directional distinction of the companion paper carried as a covector rather than as a magnitude:

$$\omega := F / (c\, u_r) = -\ \nabla u_r / (3\, \kappa_t\, u_r).$$

Proposition 9.1. With ω as above, the curvature is $\nabla \times \omega = (\nabla\kappa_t \times \nabla u_r) / (3\, \kappa_t^2\, u_r)$, so $H = d\omega$ vanishes if and only if $\nabla\kappa_t$ is parallel to ∇u_r. In particular it vanishes identically whenever the medium is homogeneous.

$$\nabla \times \omega = (\ \nabla\kappa_t \times \ \nabla u_r\,) / (\, 3\, \kappa_t^2\, u_r\,).$$

Proof. Write $\omega = -f\nabla u_r$ with $f = 1/(3\kappa_t u_r)$. Then $\nabla\times\omega = -\nabla f \times \nabla u_r$, since the curl of a gradient vanishes. Computing $\nabla f = -(u_r\nabla\kappa_t + \kappa_t\nabla u_r)/(3\kappa_t^2 u_r^2)$ and taking the cross product with ∇u_r annihilates the second term, leaving $\nabla\times\omega = (\nabla\kappa_t \times \nabla u_r)/(3\kappa_t^2 u_r)$. The identity was verified symbolically. ■

A homogeneous scattering medium is flat. The sector gravitates only where the structure of the medium and the structure of the radiation field are misaligned.

This is stronger and more restrictive than the loose expectation it replaces, and it corrects it. It is not enough that a medium scatters. Scattering supplies the mechanism by which differences fail to compose, and misalignment of the two gradients supplies the curvature. In a uniform fog, however optically thick, the sector carries no local source. At the edge of a cloud, where the opacity gradient runs across the illumination gradient, it does.

Corollary 9.2 (Source criterion, radiative form). In the diffusion regime the co-distinction sector supplies a non-vanishing local stress-energy if and only if the two gradients are non-parallel. Homogeneous media, and media whose opacity is stratified parallel to the radiation gradient, are gravitationally inert in this sector.

> Proposition 9.1 is PROVED within the diffusion closure and was checked symbolically. Its restriction to that closure is a real limitation: outside the diffusion regime the transport one-form is not the normalised flux and the computation must be redone. What is settled is that the question has a definite answer for a named kernel, and that the answer is a misalignment condition rather than a blanket yes.

PART IV · THE WEAK FIELD

10. The bridge, stated

Expand about weak, slowly varying fields, $g_{00} \approx -(1 + 2\Phi\_N/c^2)$ and $g\_ij \approx (1 - 2\Psi/c^2)\delta\_ij$. The 00 component reduces under the usual non-relativistic assumptions to

$$\nabla^2\Phi\_N = 4\pi G\,\rho\_eff, \qquad g\_N = -\nabla\Phi\_N, \qquad \mu\_eff = G\,M\_eff = G\textstyle\int\_\Sigma\,(T_{00}{}^\wedge UF/c^2)\sqrt{\gamma}\,d^3x.$$

This is the bridge the visual programme requires. The rendered φ is not automatically the Newtonian potential, and a model must either derive a calibration $\Phi\_N = f(\varphi,\psi,\mathcal{D})$ or show that φ itself obeys the correct weak-field equation with the correct source and dimensions, without inserting the desired result by hand. The next section does the second of these, and finds that it holds in a regime rather than identically.

Once the bridge is fixed the field maps become predictions rather than reorganisations, since position and velocity come from observation while the field acceleration comes from solved equations. The decisive quantity is then the residual g_observed − g_UF across Solar System ephemerides, binaries, exoplanets, lensing and larger structures.

11. How far the identification of φ with the Newtonian potential can be taken

The bridge can be examined rather than deferred. In the static limit with K constant the scalar equation of motion reads $K\nabla^2\varphi = V'(\varphi) + \partial U/\partial\varphi$, while the Newtonian potential obeys $\nabla^2\Phi\_N =$

$4\pi G\rho\_eff$. Identifying the two requires the source of the scalar equation to be proportional to the total effective density.

It cannot be so identically, and the obstruction is specific. The effective density contains the scalar own gradient energy,

$$\rho\_eff = [\ \tfrac{1}{2}K|\nabla\varphi|^2 + V(\varphi) + |D_0\psi|^2/c^2 + |D\_i\psi|^2 + U + \rho\_rad\ c^2 + \rho\_active\ c^2\ ]\ /\ c^2,$$

so the term $\tfrac{1}{2}K|\nabla\varphi|^2$ makes φ a source of its own equation, whereas the Poisson source must be independent of Φ_N for the identification to be linear.

Proposition 11.1. The identification of φ with Φ_N holds at first post-Newtonian order, with fractional corrections of order Φ_N/c^2 arising from the scalar own gradient energy in ρ_eff. It is exact only in the test-field limit, where $\tfrac{1}{2}K|\nabla\varphi|^2$ is negligible against the matter terms.

This is not a defect peculiar to the construction. It is the ordinary nonlinearity of gravity, which sources itself, appearing in the scalar sector. What it fixes is the size of the effect, and therefore where the bridge can be trusted:

Location	Φ_N/c^2	Status of the identification
Solar surface	2.1×10^{-6}	good to one part in 5×10^5
Earth orbit	1.0×10^{-8}	good to one part in 10^8
Neutron star surface	about 2×10^{-1}	fails; full solution required

So the Solar System and exoplanet maps sit in the regime where the identification is safe to well beyond their own observational precision, and the strong-field tests do not. That divides the falsification programme cleanly, and it removes the bridge from the list of things that must be settled before any comparison can be made.

> Proposition 11.1 is PROVED at the stated order and is standard post-Newtonian bookkeeping. It closes the weak-field item in the regime where the visual programme operates. It does not close the strong-field case, which requires the isolated-source metric.

PART V · FALSIFICATION

12. The programme, in order of what kills fastest

	Task	Status after this paper
1	Covariant conservation of the source	CLOSED. Automatic by Proposition 5.1; not a per-model check.

	Task	Status after this paper
2	Curvature of the co-distinction sector for a named kernel	CLOSED in the diffusion closure by Proposition 9.1. The criterion is misalignment of the two gradients.
3	Weak-field identification of φ with Φ_N	CLOSED to 1PN by Proposition 11.1, with the correction order and its size stated. Strong field remains open.
4	Prediction of μ from closure structure	OPEN, and forced by Theorem 1.1. Until this exists no orbital data tests the sector.
5	Finite-energy localised solutions of the chosen action	OPEN. No closures, no sources.
6	Isolated-source metric; exterior against Schwarzschild; PPN parameters	OPEN. The strong-field half of item 3.
7	Solar System proxy replaced by Horizons state vectors	OPEN, but now meaningful, since item 3 licenses the comparison at that precision.

Items 1 to 3 were listed as programme tasks in the earlier draft and are settled above. What that leaves is a shorter and sharper list, and item 4 is the one that decides whether any of it is testable.

Item 4 is forced by Theorem 1.1. Until μ is predicted from structure rather than read off orbits, no quantity of orbital data can test the source sector, because the data identify only the number the model was handed. That is not a difficulty peculiar to this framework. It is what the identifiability theorem says about any theory of gravitational sources.

13. Status ledger

Statement	Status	Note
Orbits identify μ and not G or M separately	PROVED	Theorem 1.1
The μ quotient is a redundancy, not a loss	PROVED	fibres stable under the dynamics
Locally potentiated implies flat implies zero source	PROVED	Proposition 8.1
Local source iff cocycle fails at small triples	PROVED	Corollary 8.2
Curvature equals the gradient misalignment	PROVED in diffusion	Proposition 9.1, checked symbolically
Homogeneous scattering media are inert	PROVED in diffusion	Corollary 9.2
Covariant conservation of the source	PROVED	Proposition 5.1, Noether second theorem

Statement	Status	Note
φ equals Φ_N at 1PN, error of order $Φ\_N/c^2$	PROVED	Proposition 11.1
Candidate action and T_μν	CANDIDATE	not derived from the axioms
Curvature outside the diffusion regime	OPEN	the one-form is not the normalised flux there
Strong-field bridge	OPEN	requires the isolated-source metric
Prediction of μ from structure	OPEN	required by Theorem 1.1
Derivation of SI G	not attempted	needs an independent inertial standard

ABOVE THIS LINE: STANDARD GENERAL RELATIVITY AND FIELD THEORY
BELOW THIS LINE: THE AUTHOR'S OWN FRAMEWORK, OFFERED AS INTERPRETATION

PART VI · EXTENSION · THE FIFTH DEGENERACY

14. The line, and what is below it

Everything above is standard general relativity and standard field theory. Theorem 1.1 is two lines of substitution. Proposition 8.1 is $d^2 = 0$. The action is a candidate written in the usual way. A reader who rejects what follows keeps all of it.

What follows uses the Corpus grades: PROVED, CONDITIONAL, PROPOSED, OPEN.

15. Five degeneracies, and two kinds

This programme has now met the same structure five times, and the fifth was found without looking for it.

Where	What is identified	Kind
Gödel closure	A_0 with A_1, by the projection	loss: traversal does not descend
Bulk action class	Einstein–Hilbert with Γ–Γ	loss: boundary data does not descend

Where	What is identified	Kind
Gauge freedom	potentials differing by a gradient	redundancy: nothing is lost
Co-distinction	all $\mathcal{D}$ with the same potential	loss iff the cocycle fails at small triples
Orbital data	(G,M) with (λG, M/λ)	redundancy: nothing is lost

The programme's earlier papers drew the distinction between a projection that discards something dynamically relevant and one that discards only redundancy, and used the Maxwell gauge case as the control. Theorem 1.1 supplies a second control, and it is a better one, because it is a degeneracy in observation rather than in description. Nothing is lost when orbits are summarised by μ, and everything about the source normalisation is lost with it. Both statements are true, and the second is not a complaint about the first.

The same question every time: does the quotient descend? Three times it did not,
and twice it did. The programme is the habit of asking.

16. What the framework adds here

Two things, and one of them is a restriction rather than a licence.

The restriction is Proposition 8.1 together with Proposition 9.1, and it is the framework constraining itself. A bilocal primitive was introduced because differences were held to be prior to magnitudes. The propositions say that where the difference is locally a magnitude after all, the primitive has no gravitational content, and that in the one regime where the curvature has been computed the condition for having any is narrow: the two gradients must be misaligned. The framework does not get to claim it both ways. That is the kind of result a programme should want, because it removes a place where an untestable sector could have been asserted, and it names in advance the media where the sector must show nothing.

The licence is Theorem 1.1 read against the framework's own aims. If orbital data cannot separate G from M, then a framework claiming to derive a source scale is not competing with those data and cannot be checked by them. It must predict μ from closure structure first, and only then is the ensemble work a test rather than a fit. Item 7 of Section 10 states that, and it is the framework's own condition of defeat in this sector.

> Proposition 8.1 and its corollaries are PROVED and are not framework-dependent. The reading of the five degeneracies as instances of one question is PROPOSED, and it inherits the identification named in the companion papers rather than adding a new one.

17. Closing

The source sector was the place where this programme was most exposed, because a field defined on pairs has no obvious right to curve anything. It turns out to have that right under a condition, and the condition is sharp: the differences must fail to compose in the small, not merely around the world.

The other half of the exposure was empirical, and Theorem 1.1 says how it must be met. Orbits will never adjudicate between a coupling and a mass. They will adjudicate a predicted μ against a measured one, and only once the prediction exists.

A theory earns a source by saying what would leave it without one. Four such statements are proved here, and the sharpest of them says that a uniform fog carries no gravity at all in this sector, however thick it is.

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